

Physical Chemistry (I)

Lecture 19

Real Solutions



In the Last Class

- Multi-component systems
- Mixing of gases
 - Mixing of perfect gases
 - Mixing of real gases
- Mixing of liquids and solids
 - Ideal solutions



Multi-Component System

- State: $(p, T, n) \rightarrow (p, T, n_A, n_B, \dots)$
- Thermodynamic potentials
 - $dU = TdS - pdV + \mu_A dn_A + \mu_B dn_B + \dots$
 - $dH = TdS + Vdp + \mu_A dn_A + \mu_B dn_B + \dots$
 - $dA = -SdT - pdV + \mu_A dn_A + \mu_B dn_B + \dots$
 - $dG = -SdT + Vdp + \mu_A dn_A + \mu_B dn_B + \dots$

Last class

Phase Diagram

- Mole fraction of species J

$$x_J = \frac{n_J}{\sum_J n_J}$$

- At least one axis (usually the x-axis) is for x_J
- The other is for $T, p, \dots$



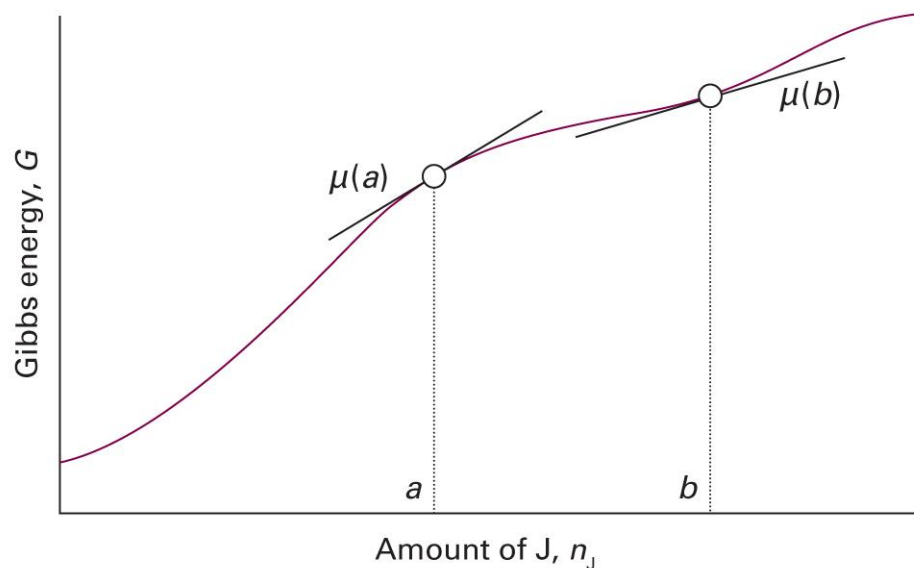
Partial Molar Quantities

$$X_J = \left(\frac{\partial X}{\partial n_J} \right)_{p,T,n'}$$

contribution that a component of a mixture makes
to the total quantity of a sample

Partial Gibbs Energy

$$G_J = \left(\frac{\partial G}{\partial n_J} \right)_{p,T,n'} = \mu_J$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.4)

Last class

Gibbs-Duhem Equation

$$SdT - Vdp + \sum_J n_J d\mu_J = 0$$

At constant T and p ,

$$\sum_J n_J d\mu_J = 0$$

Last class

Dalton's Law

Partial pressure for species J is

$$p_J = x_J p$$

so

$$p_A + p_B + \cdots = (x_A + x_B + \cdots)p = p$$

Valid for all gases (not only perfect)

Mixing of Perfect Gases

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B)$$

- Entropy of mixing

$$\Delta_{\text{mix}}S = -\left(\frac{\partial \Delta_{\text{mix}}G}{\partial T}\right)_p = -nR(x_A \ln x_A + x_B \ln x_B)$$

- Enthalpy of mixing

$$\Delta_{\text{mix}}H = \Delta_{\text{mix}}G + T\Delta_{\text{mix}}S = 0$$

Last class

Real Gases

For a perfect gas,

$$\mu = \mu^\ominus + RT \ln \frac{p}{p^\ominus}$$

For a real gas,

$$\mu = \mu^\ominus + RT \ln \frac{f}{p^\ominus}$$

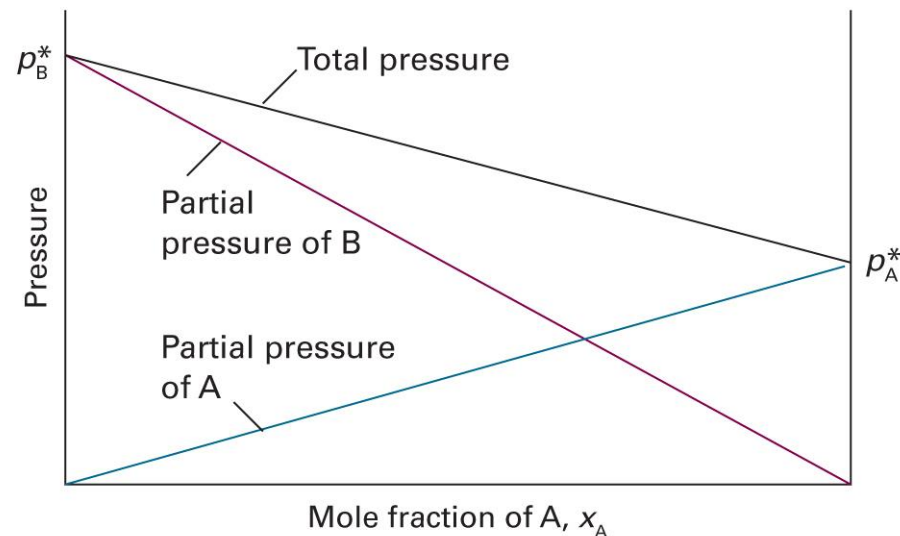
where f is the **fugacity**.



Raoult's Law: Ideal Solution

$$p_A = x_A p_A^*$$

- **Ideal solutions:** mixtures that obey the law throughout the composition range from pure A to pure B



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.11)

Ideal Solutions

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B)$$

- Entropy of mixing

$$\Delta_{\text{mix}}S = -\left(\frac{\partial \Delta_{\text{mix}}G}{\partial T}\right)_p = -nR(x_A \ln x_A + x_B \ln x_B)$$

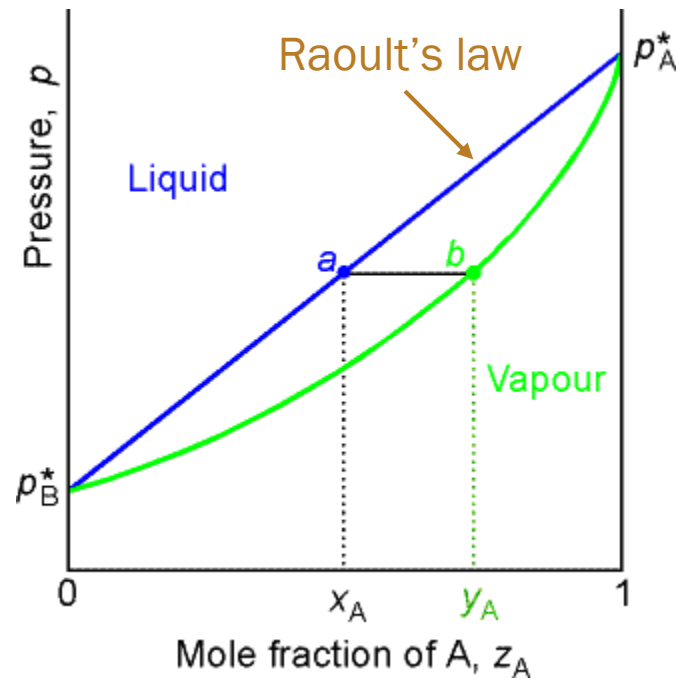
- Enthalpy of mixing

$$\Delta_{\text{mix}}H = \Delta_{\text{mix}}G + T\Delta_{\text{mix}}S = 0$$

(Same with perfect gas mixtures)

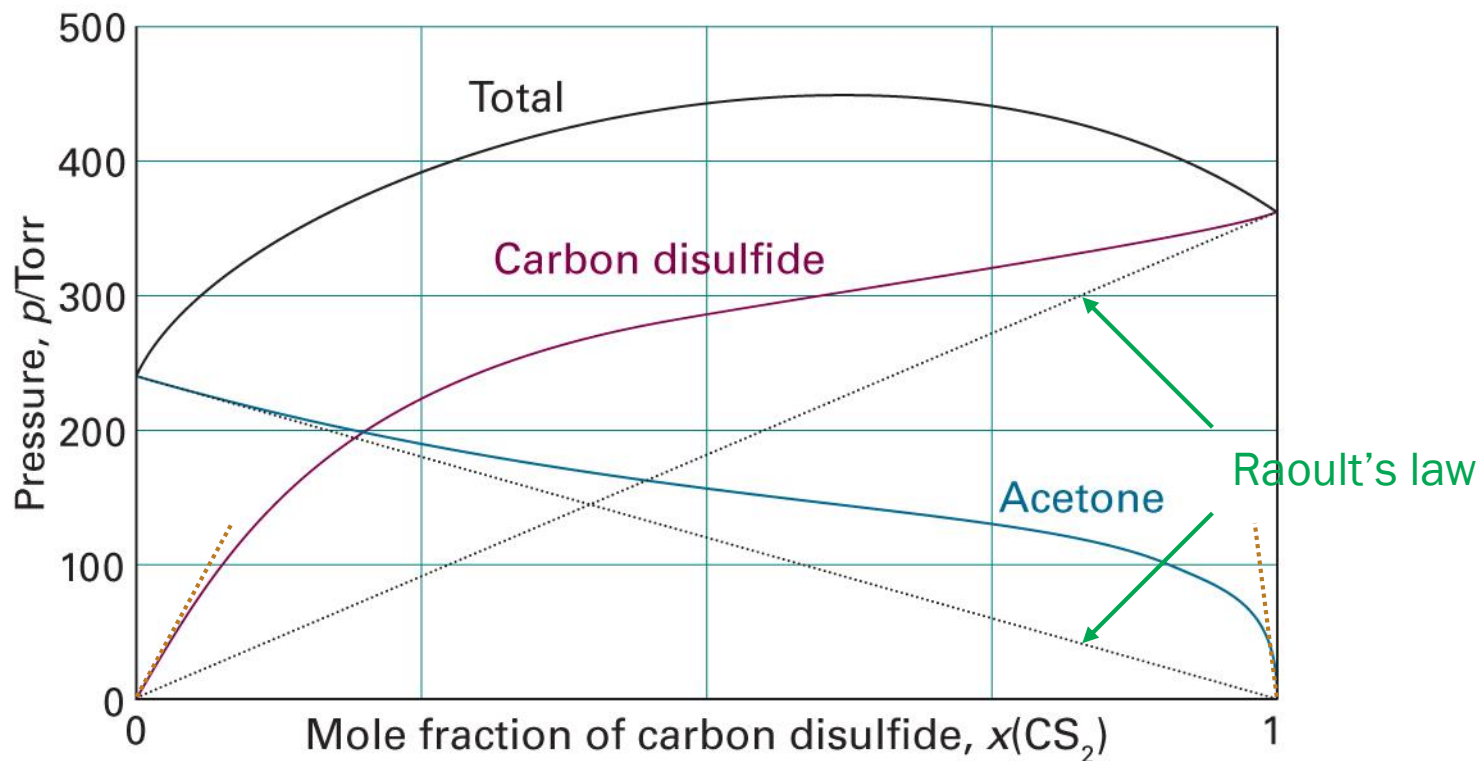
Last class

Pressure-Composition Diagram



(Image: <http://people.cst.cmich.edu/teckl1mm/pchemi/chm351ch8af01.htm>)

Real Solutions: Deviations



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.12)

Henry's Law

$$p_B = x_B K_B$$

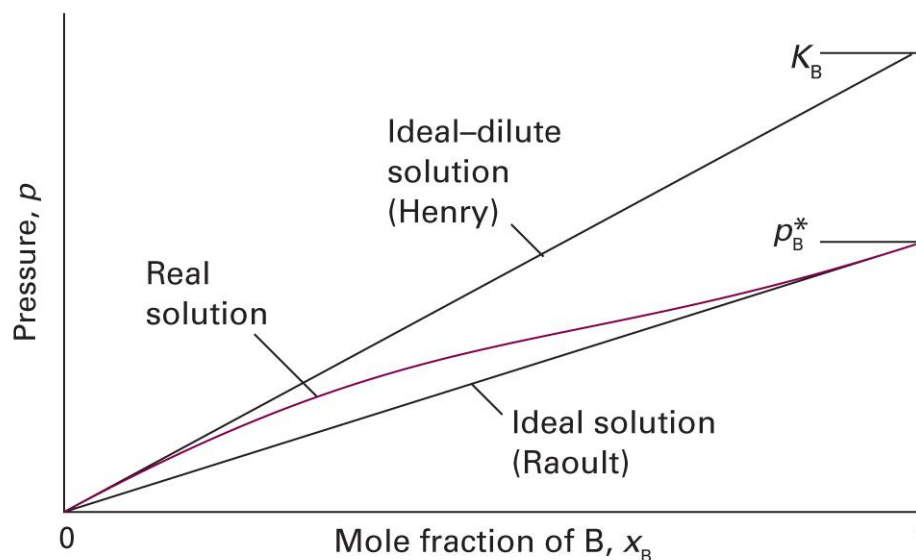
- **Ideal-dilute solutions:** mixtures for which the solute B obeys Henry's law and the solvent A obeys Raoult's law

The molality, b , is often instead used:

$$p_B = b_B K_B$$

Limiting Laws for Real Solutions

- Raoult's law: reliable as $x \rightarrow 1$
- Henry's law: reliable as $x \rightarrow 0$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.13)

μ for Ideal-Dilute Solutions

- Solvent: Raoult's law

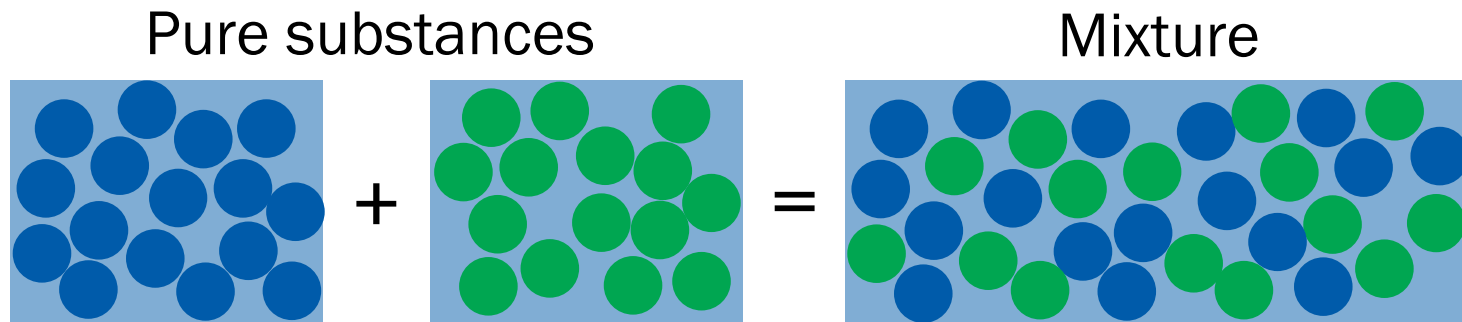
$$\mu_A(l) = \mu_A^*(l) + RT \ln(p_A/p_A^*) = \mu_A^*(l) + RT \ln x_A$$

- Solute: Henry's law

$$\begin{aligned}\mu_B(l) &= \mu_B^*(l) + RT \ln(p_B/p_B^*) = \mu_B^*(l) + RT \ln(K_B x_B/p_B^*) \\ &= \underbrace{\mu_B^*(l) + RT \ln(K_B/p_B^*)}_{\mu_B^\ominus(l)} + RT \ln x_B \\ &= \mu_B^\ominus(l) + RT \ln x_B\end{aligned}$$

Can We Make a Better Model?

Regular Solution



A model solution with

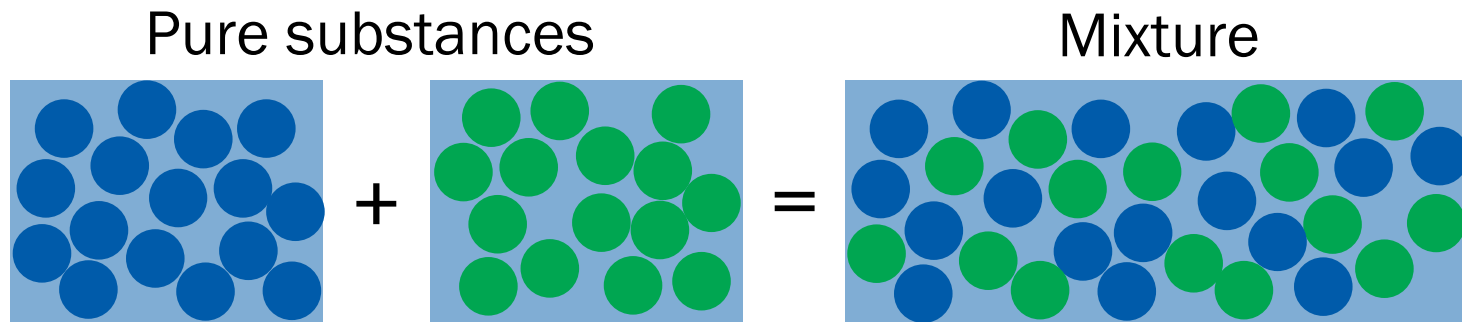
$$\Delta_{\text{mix}}S = \Delta_{\text{mix}}S^{\text{ideal}} = -nR(x_A \ln x_A + x_B \ln x_B)$$

and

$$\Delta_{\text{mix}}H \neq 0$$

Not in Atkins

Microscopic Model

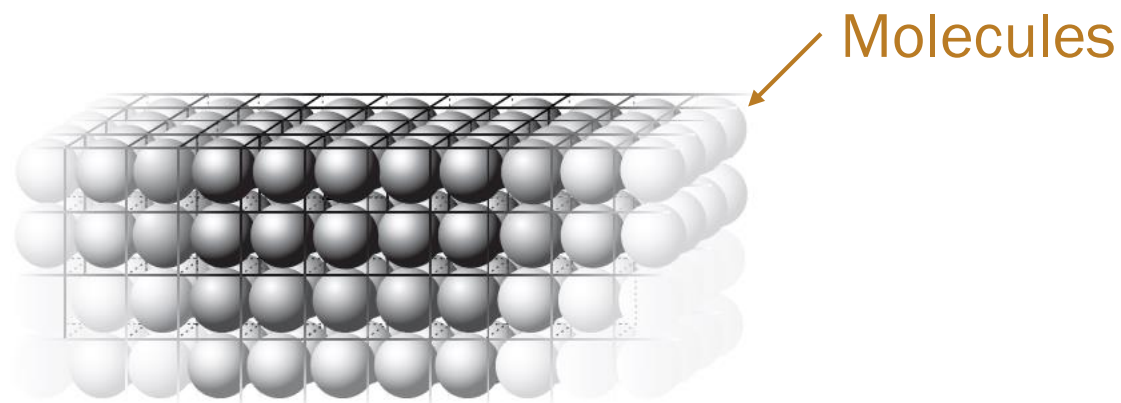


	Pure substance		Mixture
Number of A molecules	N_A	0	N_A
Number of B molecules	0	N_B	N_B
Total number of molecules	N_A	N_B	$N_A + N_B$

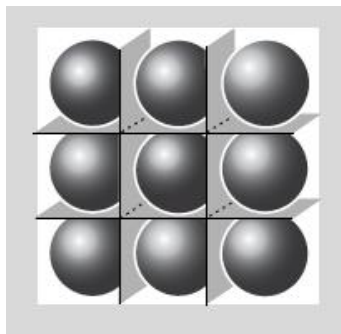
Calculate the chemical potentials

Not in Atkins

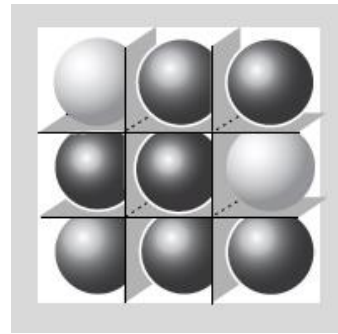
Lattice Model



Pure substance

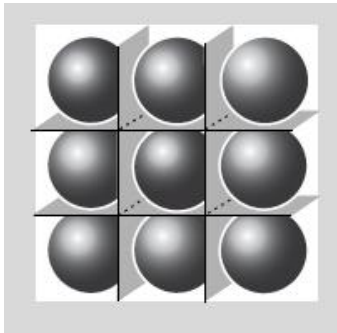


Mixture



Numbers of Microstates

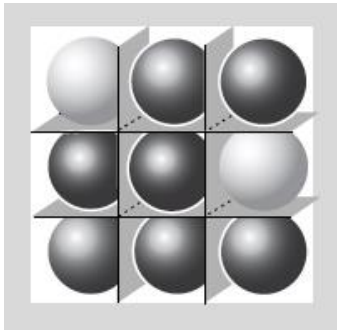
Pure



- Number of rooms: N
- Number of molecules: N

$$W = 1$$

Mixed



- Number of rooms: $N_A + N_B$
- Number of A molecules: N_A
- Number of B molecules: N_B

$$W = {}_{N_A+N_B}C_{N_A} = (N_A + N_B)! / (N_A! N_B!)$$

Entropy of Mixing

- Pure A: $S_A = k_B \ln W_A = k_B \ln 1 = 0$
- Pure B: $S_B = k_B \ln W_B = k_B \ln 1 = 0$
- Mixture:

$$S_{A+B} = k_B \ln W_{A+B} = k_B \ln \frac{(N_A + N_B)!}{N_A! N_B!}$$

- Entropy of mixing

$$\Delta_{\text{mix}} S = S_{A+B} - S_A - S_B = k_B \ln \frac{(N_A + N_B)!}{N_A! N_B!}$$

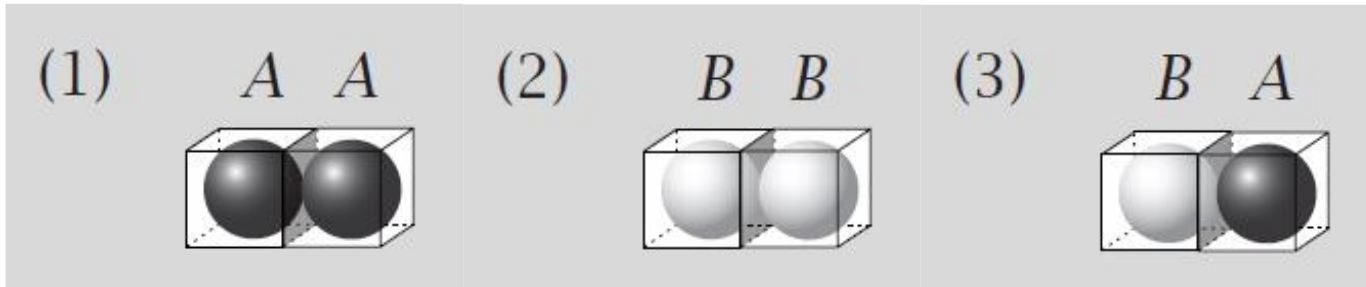
Entropy of Mixing

$$\Delta_{\text{mix}}S = k_B \ln \frac{(N_A + N_B)!}{N_A! N_B!}$$

Applying Stirling's approximation,

$$\begin{aligned}\Delta_{\text{mix}}S &= k_B \{\ln(N_A + N_B)! - \ln N_A! - \ln N_B!\} \\ &= k_B \{(N_A + N_B) \ln(N_A + N_B) - N_A \ln N_A - N_B \ln N_B\} \\ &= -k_B \left(N_A \ln \frac{N_A}{N_A + N_B} + N_B \ln \frac{N_B}{N_A + N_B} \right) \\ &= -Nk_B (x_A \ln x_A + x_B \ln x_B) \\ &= -nR(x_A \ln x_A + x_B \ln x_B): \text{ideal solution entropy!}\end{aligned}$$

Energy of Mixture



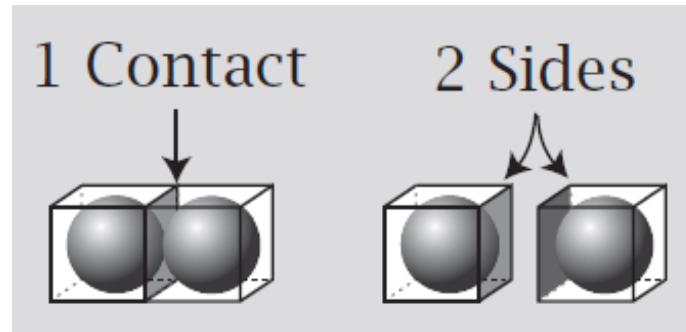
Considering three types of contacts,

$$U = m_{AA}w_{AA} + m_{BB}w_{BB} + m_{AB}w_{AB}$$

where m_{ij} is the number of i - j contacts, and

w_{ij} is the energy of each i - j contact

Contacts and Sides



- Each contact contributes 2 sides
- Each molecule contributes z sides
- Total number of “A” sides = $zN_A = 2m_{AA} + m_{AB}$
- Total number of “B” sides = $zN_B = 2m_{BB} + m_{AB}$

Energy of Mixture

$$zN_A = 2m_{AA} + m_{AB}, \quad zN_B = 2m_{BB} + m_{AB}$$

Rearrangement leads to

$$m_{AA} = (zN_A - m_{AB})/2$$

$$m_{BB} = (zN_B - m_{AB})/2$$

which gives

$$U = \frac{zN_A - m_{AB}}{2} w_{AA} + \frac{zN_B - m_{AB}}{2} w_{BB} + m_{AB} w_{AB}$$

$$U = \frac{zw_{AA}}{2} N_A + \frac{zw_{BB}}{2} N_B + \left(w_{AB} - \frac{w_{AA} + w_{BB}}{2} \right) m_{AB}$$

What is m_{AB} ?

The probability to have B in any site:

$$p_B = N_B/N$$

The average number of AB contacts that each A makes:

$$zp_B = zN_B/N$$

The total number of AB contacts:

$$m_{AB} \approx N_A \times (zN_B/N) = zN_A N_B/N$$

Energy of Mixture

$$U = \frac{z w_{AA}}{2} N_A + \frac{z w_{BB}}{2} N_B + \left(w_{AB} - \frac{w_{AA} + w_{BB}}{2} \right) \frac{z N_A N_B}{N}$$

$$U = \frac{z w_{AA}}{2} N_A + \frac{z w_{BB}}{2} N_B + \xi k_B T \frac{N_A N_B}{N}$$

where the **exchange parameter** ξ is

$$\xi = \frac{z}{k_B T} \left(w_{AB} - \frac{w_{AA} + w_{BB}}{2} \right)$$

Energy of Mixing

$$U = \frac{ZW_{AA}}{2} N_A + \frac{ZW_{BB}}{2} N_B + \xi k_B T \frac{N_A N_B}{N}$$

- Pure A: $N_B = 0$

$$U_A = \frac{ZW_{AA}}{2} N_A$$

- Pure B: $N_A = 0$

$$U_B = \frac{ZW_{BB}}{2} N_B$$

$$\Delta_{\text{mix}} U = U_{A+B} - U_A - U_B = \xi k_B T \frac{N_A N_B}{N}$$

Not in Atkins

Helmholtz Energy of Mixing

$$\Delta_{\text{mix}}S = -nR(x_A \ln x_A + x_B \ln x_B)$$

$$\Delta_{\text{mix}}U = \xi k_B T \frac{N_A N_B}{N} = \xi N k_B T x_A x_B = \xi n R T x_A x_B$$

$$\Delta_{\text{mix}}A = \Delta_{\text{mix}}U - T \Delta_{\text{mix}}S$$

$$\Delta_{\text{mix}}A = nRT(x_A \ln x_A + x_B \ln x_B + \xi x_A x_B)$$

Energy of Mixing

For a regular solution,

$$\Delta_{\text{mix}}U = nRT\xi x_A x_B$$

$$\xi = \frac{z}{k_B T} \left(w_{AB} - \frac{w_{AA} + w_{BB}}{2} \right)$$

- $\xi > 0$, or $w_{AB} > (w_{AA} + w_{BB})/2$: mixing is endothermic
- $\xi < 0$, or $w_{AB} < (w_{AA} + w_{BB})/2$: mixing is exothermic
- $\xi = 0$, or $w_{AB} = (w_{AA} + w_{BB})/2$: ideal solution

Energy to Enthalpy

For liquid-liquid mixing,

$$\Delta_{\text{mix}}H = H_{\text{A+B}} - H_{\text{A}} - H_{\text{B}}$$

$$H_{\text{A+B}} = U_{\text{A+B}} + (pV)_{\text{A+B}}$$

$$H_{\text{A}} = U_{\text{A}} + (pV)_{\text{A}}$$

$$H_{\text{B}} = U_{\text{B}} + (pV)_{\text{B}}$$

Hence,

$$\Delta_{\text{mix}}H = \Delta_{\text{mix}}U + (pV)_{\text{A+B}} - (pV)_{\text{A}} - (pV)_{\text{B}} \approx \Delta_{\text{mix}}U$$

$$\Delta_{\text{mix}}G = \Delta_{\text{mix}}H - T\Delta_{\text{mix}}S \approx \Delta_{\text{mix}}U - T\Delta_{\text{mix}}S = \Delta_{\text{mix}}A$$

Model Summary

- Ideal solution

- Raoult's law: $p_A = x_A p_A^*$
- $\Delta_{\text{mix}}G = -T\Delta_{\text{mix}}S = nRT(x_A \ln x_A + x_B \ln x_B)$

- Ideal-dilute solution

- Solvent – Raoult's law
- Solute – Henry's law: $p_B = x_B K_B$

- Regular solution

- $\Delta_{\text{mix}}G = \Delta_{\text{mix}}H - T\Delta_{\text{mix}}S^{\text{ideal}}$
- Lattice model (Bragg-Williams model): $\Delta_{\text{mix}}H = nRT\zeta x_A x_B$

Deviations from Ideality

- Excess function

$$\chi^E = \Delta_{\text{mix}}X - \Delta_{\text{mix}}X^{\text{ideal}}$$

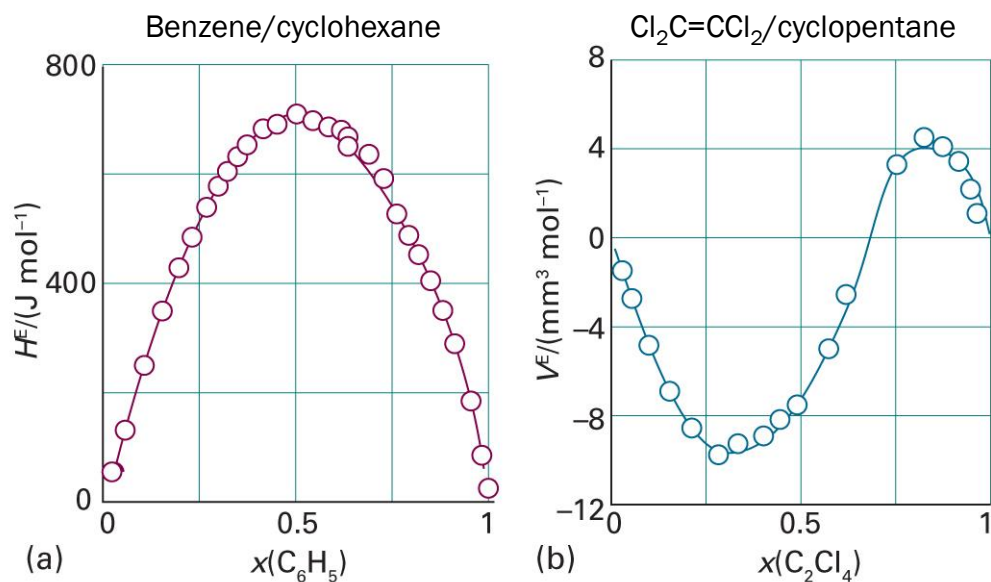
- Activity

$$\mu_A = \mu_A^* + RT \ln a_A$$

Excess Function

$$X^E = \Delta_{\text{mix}}X - \Delta_{\text{mix}}X^{\text{ideal}}$$

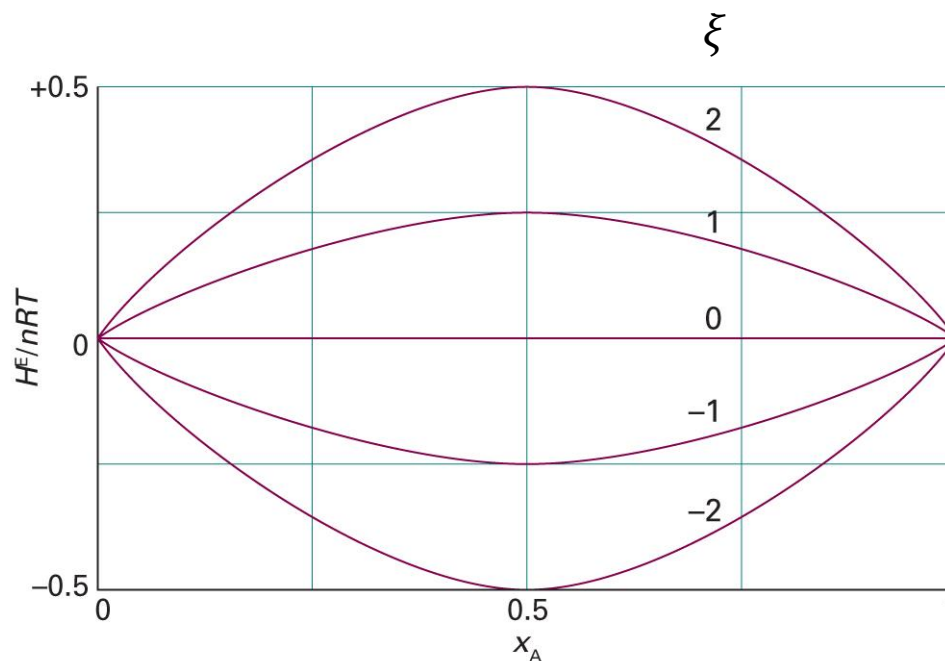
- Regular solution: $S^E = 0$ and $H^E \neq 0$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5B.3)

BW Model: Excess Enthalpy

$$H^E = \xi nRT x_A x_B$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5B.4)

Activity a_J

- Ideal-dilute solution

- Solvent: $\mu_A(l) = \mu_A^*(l) + RT \ln x_A$

- Solute: $\mu_B(l) = \mu_B^\ominus(l) + RT \ln x_B$

where $\mu_B^\ominus(l) = \mu_B^*(l) + RT \ln(K_B/p_B^*)$

- Real solution

- Solvent: $\mu_A(l) = \mu_A^*(l) + RT \ln a_A$

- Solute: $\mu_B(l) = \mu_B^\ominus(l) + RT \ln a_B$

Determination of Activity

$$\mu(l) = \mu^*(l) + RT \ln(p/p^*) \text{ (perfect gas)}$$

- Solvent

$$\mu_A(l) = \mu_A^*(l) + RT \ln a_A$$

$$a_A = p_A/p_A^*$$

- Solute

$$\mu_B(l) = \mu_B^\ominus(l) + RT \ln a_B = \mu_B^*(l) + RT \ln(K_B a_B/p_B^*)$$

$$a_B = p_B/K_B$$

Activity Coefficient

$$\gamma_J = a_J/x_J$$

- Solvent

$$\mu_A(l) = \mu_A^*(l) + RT \ln a_A = \mu_A^*(l) + RT \ln x_A + RT \ln \gamma_A$$

- Solute

$$\mu_B(l) = \mu_B^\ominus(l) + RT \ln a_B = \mu_B^\ominus(l) + RT \ln x_B + RT \ln \gamma_B$$

As $x_A \rightarrow 1$ and $x_B \rightarrow 0$, $a_J \rightarrow x_J$ and $\gamma_J \rightarrow 1$

Gibbs Energy of Mixing

$$\begin{aligned}\Delta_{\text{mix}}G &= G - G^* \\&= n_A\mu_A + n_B\mu_B - n_A\mu_A^* - n_B\mu_B^* \\&= n_A(\mu_A - \mu_A^*) + n_B(\mu_B - \mu_B^*) \\&= n_ART \ln a_A + n_BRT \ln a_B \\&= nRT(x_A \ln a_A + x_B \ln a_B) \\&= nRT(x_A \ln x_A + x_B \ln x_B + \underbrace{x_A \ln \gamma_A + x_B \ln \gamma_B}_{\text{deviation from ideality}})\end{aligned}$$

Activity of BW Model

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B + x_A \ln \gamma_A + x_B \ln \gamma_B)$$

Comparing with the BW model,

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B + \xi x_A x_B)$$

$$x_A \ln \gamma_A + x_B \ln \gamma_B = \xi x_A x_B$$

Activity of BW Model

$$x_A \ln \gamma_A + x_B \ln \gamma_B = \xi x_A x_B$$

$$\xi x_A x_B = \xi x_A x_B (x_A + x_B) = x_A (\xi x_B^2) + x_B (\xi x_A^2)$$

The following equations satisfy the given criterion:

$$\ln \gamma_A = \xi x_B^2, \text{ or } \gamma_A = e^{\xi x_B^2}$$

$$\ln \gamma_B = \xi x_A^2, \text{ or } \gamma_B = e^{\xi x_A^2}$$

(Margules equations)

BW Model and the Two Laws

$$\gamma_A = e^{\xi x_B^2}, \quad \gamma_B = e^{\xi x_A^2}$$

For component A, the activity is

$$a_A = x_A \gamma_A = x_A e^{\xi x_B^2} = x_A e^{\xi(1-x_A)^2}$$

From the property of the activity, $a_A = p_A/p_A^*$ and

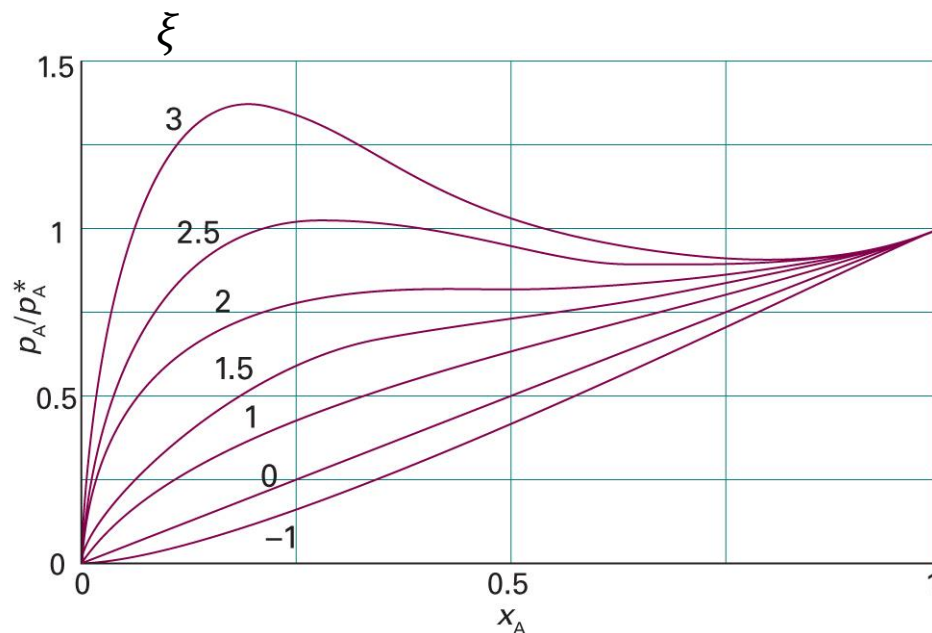
$$p_A = p_A^* a_A = p_A^* x_A e^{\xi(1-x_A)^2}$$

- $x_A \rightarrow 1$: $p_A = p_A^* x_A e^0 = p_A^* x_A$ (Raoult's law)
- $x_A \rightarrow 0$: $p_A = p_A^* x_A e^{\xi} = K_A x_A$ (Henry's law)

$$\text{where } K_A = p_A^* e^{\xi}$$

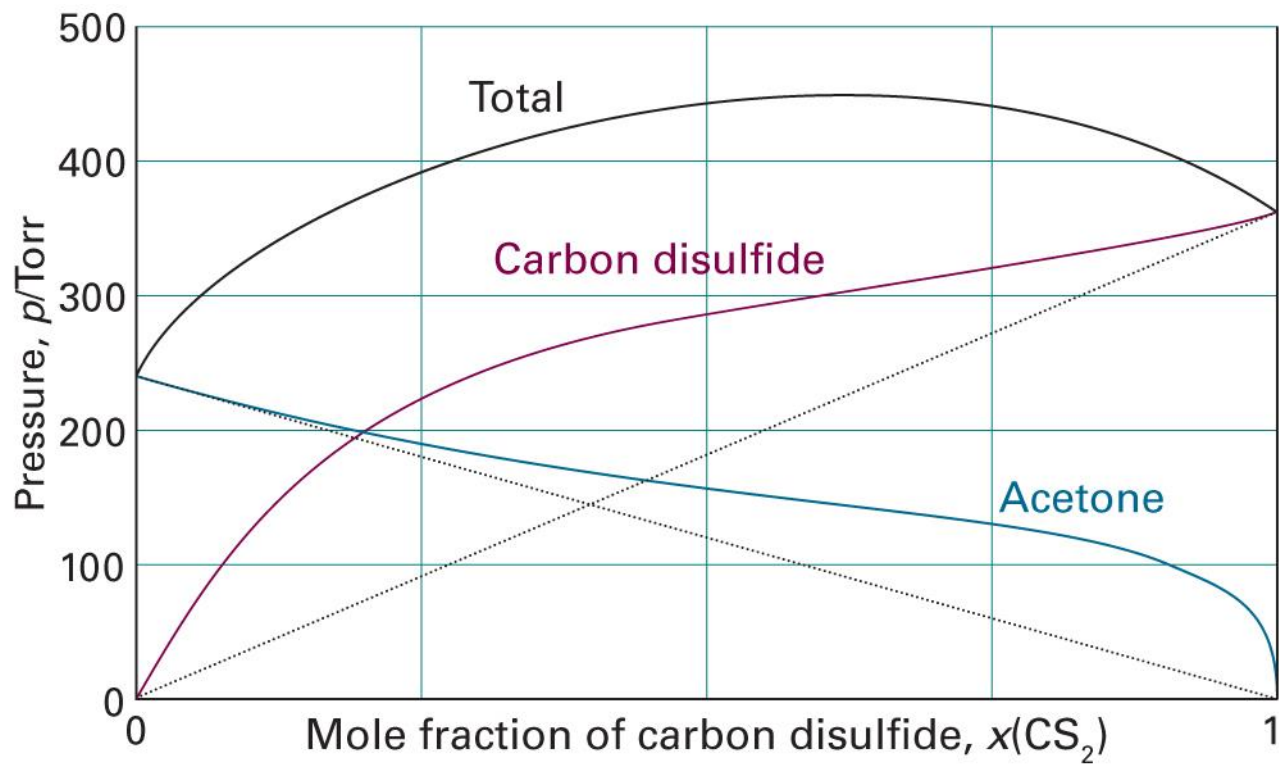
Vapor Pressure of BW Model

$$p_A/p_A^* = x_A e^{\xi(1-x_A)^2}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5F.2)

Now We Can Explain This



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.12)